Pocket AI physics

Eleven chapters on the physics and engineering a software engineer needs to understand why the brass build is designed the way it is. Each chapter starts from the physics, builds the engineering idea on top of it with this device's real numbers, and ends with the rules it explains. The short illustrated version is Why the Brass Build Works; this is the long one. About an hour and a quarter. Any order after chapter 1.

Chapters

1. Charge, voltage, current
2. Resistance, Ohm's law, power
3. Series, parallel, and why drivers win
4. Capacitors: charge and time
5. Digital signals: thresholds and pulls
6. I²C: a bus with addresses
7. Sound into numbers: sampling and I²S
8. Making sound: class D, the bridge, decibels
9. The antenna and the near field
10. Bare conductors: when they work and when they don't
11. Heat, solder and flux

HOW TO READ IT

Chapters 1 to 4 are the foundation: charge and loops, resistance and power, dividers, capacitors. Chapter 5 turns that into digital logic. Chapters 6 to 8 are the three ways the parts talk: I²C, I²S and the amplifier. Chapter 9 is the radio. Chapters 10 and 11 are the two things specific to building it in brass: bare conductors and solder. Equations are kept to the handful you can check with a calculator against the numbers on your bench.

CHAPTER 1 OF 11

Charge, voltage, current

Metal is full of loose electrons

In brass or copper, the outer electrons don't belong to any atom. They drift around like a gas inside the metal. A circuit doesn't create charge; it pushes charge that is already there.

The push is an electric field. Pile electrons up at one end of a rod and every free electron in between feels it. The electrons themselves crawl, a tenth of a millimetre per second. The field that pushes them travels at nearly the speed of light. When A1 flips a pin, the far end of the rod knows in a fraction of a nanosecond.

Voltage is a difference, never a point

Voltage is energy per coulomb of charge. It is always *between* two points. "This pin is at 3.3 V" means 3.3 V above the reference, and the reference in this device is A1's GND pin.

Nothing is physically special about GND. It is the origin of the coordinate system, the node everyone agreed to call zero.

The brass frame is not on that coordinate system. It floats at whatever charge lands on it. That is why nothing with a defined voltage may touch it.

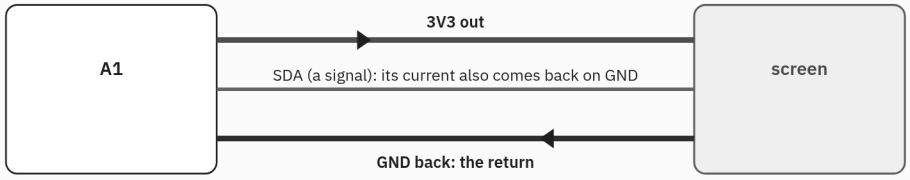
Current is charge per second

One amp is one coulomb per second. A1 idles at about 80 mA and bursts to 300 mA when the radio transmits. The amp pulls pulses of a couple of hundred milliamps at your volume. Small numbers for a house, big for a chip: 300 mA through a one-ohm bad joint drops 0.3 V, a tenth of the supply.

Two laws you need

Charge is conserved. Whatever flows into a node flows out. So current only ever flows in loops. A rod from A1 to the screen carries current only if there is a path back. That path is the ground rail, for every module and every signal at once.

Round any loop, the voltages sum to zero. What the supply pushes up, the loads drop. Those two rules are all the circuit theory in the wiring plan.



No return path, no current, no signal. Every wire in the device is half of a loop whose other half is the ground rail.

Electricity only moves in closed loops. That is the whole reason the device has one ground rail that every module touches: it is the shared return half of every loop, power and signal alike.

> IN THE BRASS BUILD, THIS IS WHY
>
> - one GND rail that every module taps: the shared return half of every loop
> - the amp's return current, the biggest, gets its own thick path so its pulses stay off the mic's return
> - the frame has no defined voltage, so nothing with one may touch it

CHAPTER 2 OF 11

Resistance, Ohm's law, power

Bumping into the lattice

Electrons pushed through a metal keep hitting the ions and dumping energy into them as heat. Voltage over current is the resistance, and for metals it is a constant. That is Ohm's law: V = I × R.

A piece of conductor's resistance comes from its material, its length and its thickness:

```
R = ρ × L / A

brass  ρ ≈ 6.4 × 10⁻⁸ Ω·m   1.0 mm rod, 40 mm:   ≈ 3.3 mΩ
copper ρ ≈ 1.7 × 10⁻⁸ Ω·m   30 AWG wire, 40 mm:  ≈ 14 mΩ
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Brass is four times worse than copper by material. Your rod has sixteen times the cross-section of 30 AWG wire. The rod wins. Either way: milliohms, a millivolt of drop at 0.3 A, nothing against 3.3 V.

So for the conductors in this device, resistance is irrelevant. It only matters where you add it on purpose.

Resistors are resistance on purpose

A resistor is a carbon film on a ceramic rod with two leads, built to a fixed value. You use one to set a current or to split a voltage. Nearly every resistor in this build does the second job; chapter 3 shows how.

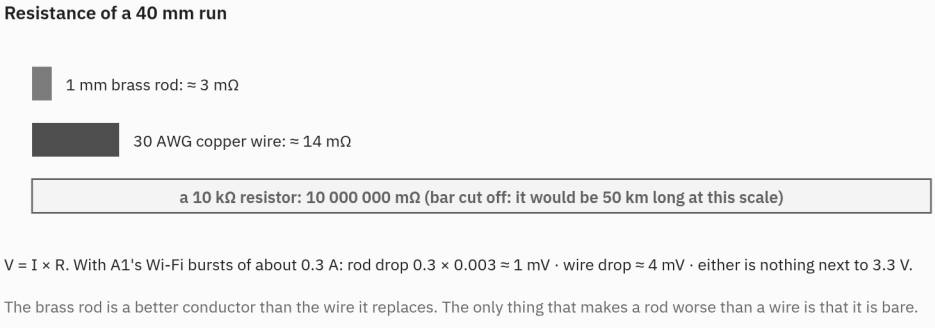
Its one rating that matters here is power: 1/4 W for yours.

Power is where heat comes from

P = V × I. With Ohm's law: P = I²R = V²/R. Every heat question in the device is one of these.

A1's regulator turns 5 V into 3.3 V by acting like a variable resistor in series with the load. It burns (5 – 3.3) × I. At 100 mA that is 0.17 W: warm, fine.

The fast-track dummy load was twelve 100 Ω resistors in parallel: 8.3 Ω. The speaker's 0.8 W splits twelve ways, 0.07 W each. One 8 Ω resistor would have cooked.



Ohm's law, V = I × R, with this device's real numbers. Conductors are near zero, resistors are deliberately huge, and the drop across a rod is a thousandth of a volt. Rods are not a compromise electrically; they only need to be kept apart.

> IN THE BRASS BUILD, THIS IS WHY
>
> - bare rod is an excellent conductor; the only cost is keeping rods apart
> - resistor values are set by the current they allow (V/R), never by the signal
> - the LDO on A1 running warm is physics, not a fault

CHAPTER 3 OF 11

Series, parallel, and why drivers win

Series adds, parallel shares

Resistances in series add; the same current goes through each and each drops a share of the voltage. Resistances in parallel share the current; the combination is smaller than any one of them.

From the series rule comes the voltage divider, the single most useful idea in this build:

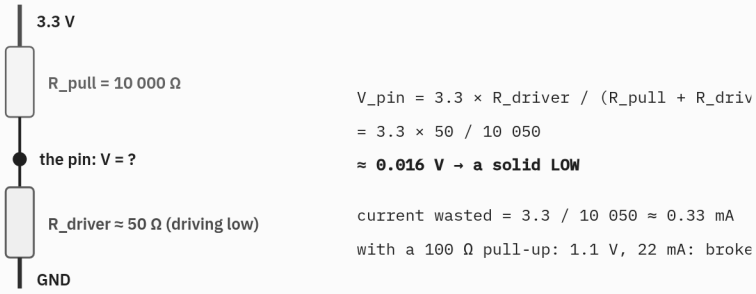
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V_middle = V_supply × R_lower / (R_upper + R_lower)
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A pull-up versus a chip driver is a divider

When a pin drives low it connects itself to ground through a few tens of ohms. Put a 10 kΩ pull-up on that pin and you have a divider: 10 kΩ on top, 50 Ω below.

The pin sits at 3.3 × 50 / 10 050 ≈ 16 mV. The driver wins two hundred to one. Cost: 0.33 mA through the pull-up while the line is low.

Make the pull-up 100 Ω and the pin sits at 1.1 V, neither high nor low, wasting 22 mA. That is why pull resistors are thousands of ohms: strong enough to beat noise, weak enough to lose to any driver.



The divider is the single most useful circuit idea in this build. Two resistances in series split a voltage in proportion to their size. A pull resistor and a chip's driver are exactly that: whichever is smaller holds the node near its own supply.

Supplies sag

Every source has internal resistance, and the USB cable adds a few hundred milliohms. A 300 mA burst drops about 0.1 V in the cable alone. A laptop port that limits current sags far more and can reset A1. Hence the fast track's advice: a wall adapter if A1 resets on loud replies.

Same thought behind the thick 5 V and ground rods for the amp: the loop carrying the most current gets the least resistance.

Why the amp gets its own ground

All the return currents flow through the ground rail's few milliohms. The amp's pulses through that shared metal make the ground wobble, and the microphone measures everything against that ground.

Give the amp its own return to A1's GND pin and the two currents meet only at the pin. Engineers call it a star ground. The splice drawing on the wiring plan is one.

> IN THE BRASS BUILD, THIS IS WHY
>
> - 10 kΩ and 100 kΩ pulls: overridable defaults
> - the amp's own thick ground and 5 V rods: the heaviest loop gets the lowest resistance
> - a wall adapter rather than a laptop port if A1 resets during loud replies

Capacitors: charge and time

Two plates, an insulator, stored charge

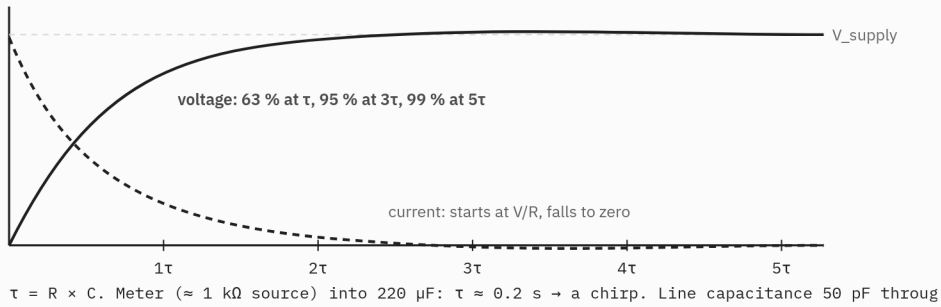
Push charge onto one plate and the opposite charge gathers on the other. The insulator stops them meeting, so a voltage builds. Charge per volt is capacitance: Q = C × V.

A farad is huge. Yours are 220 microfarads (the electrolytic, with the stripe) and 100 nanofarads (the ceramic). The 220 µF at 5 V stores ½CV² = 2.75 millijoules. Not much, but deliverable in an instant.

Charging takes time: τ = R × C

Through a resistor, a capacitor fills along an exponential curve. 63 % full at one time constant, 95 % at three, done at five.

Your meter's continuity mode pushes a small current from about a kilohm. Into 220 µF that is τ ≈ 0.2 s: a chirp, then silence as the capacitor fills. A steady tone means the current never stopped. That is a short.



Charging is exponential, and the time constant is just R × C. The same curve explains why the meter chirps across a capacitor, why a pull-up limits how fast a bus can run, and why the 220 µF can only cover the first moments of a current spike.

A capacitor passes change and blocks steady state

Its opposition to AC falls with frequency: X = 1 / (2πfC). The 220 µF is 0.7 Ω at 1 kHz and 7 mΩ at 100 kHz on paper.

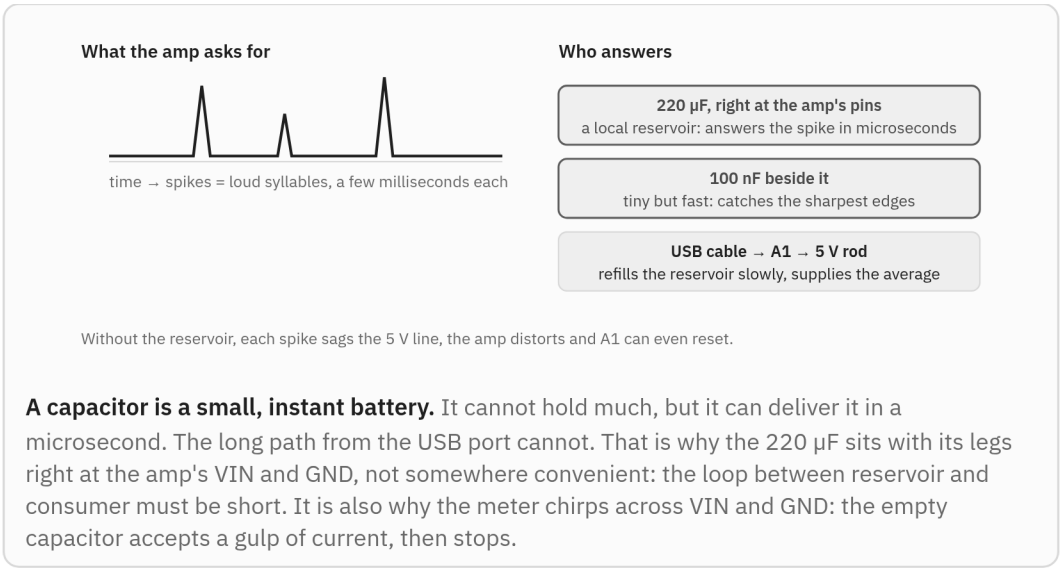
Real electrolytics have internal resistance and inductance; above a hundred kilohertz or so they stop acting like capacitors. The 100 nF ceramic beside it is useless for bulk storage and nearly perfect to tens of megahertz. The pair covers bass pulses and switching edges.

Why it must sit at the amp's pins

Every conductor is also an inductor, about a nanohenry per millimetre, and an inductor fights changes in current: V = L × di/dt.

USB port to amp is maybe 300 mm: 300 nH. A class-D amp changes current in tens of nanoseconds. 0.3 A in 30 ns through 300 nH drops 3 V. The supply at the amp would collapse on every edge.

A capacitor a few millimetres from the pins makes the loop a few nanohenries and feeds the edge from its own store. The long path only has to carry the average. Decoupling is about loop length first, capacitance second.



A capacitor is a small, instant battery. It cannot hold much, but it can deliver it in a microsecond. The long path from the USB port cannot. That is why the 220 µF sits with its legs right at the amp's VIN and GND, not somewhere convenient: the loop between reservoir and consumer must be short. It is also why the meter chirps across VIN and GND: the empty capacitor accepts a gulp of current, then stops.

> IN THE BRASS BUILD, THIS IS WHY
>
> - 220 µF + 100 nF with the shortest leads at the amp's VIN and GND
> - stripe to GND on the electrolytic; backwards, it heats and can burst
> - the meter chirps then stops across VIN and GND; a steady tone is a fault

CHAPTER 5 OF 11

Digital signals: thresholds and pulls

A pin carries a voltage, not a bit

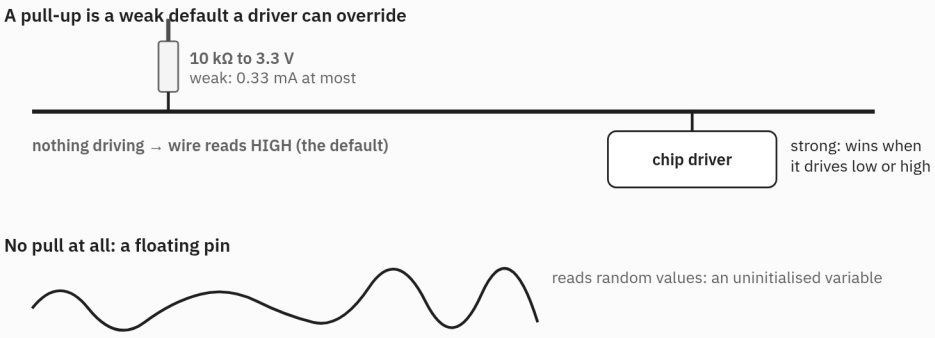
The ESP32-C3 reads anything under about 0.8 V as low and anything over 2.5 V as high. In between is a forbidden band where the input may read either, flicker, or draw extra current.

A well-driven line sits near 0 or near 3.3 V with a volt of margin each side. That margin is why digital circuits shrug off noise that would wreck an analogue one.

An input left alone floats

An input pin is gigaohms, drawing nanoamps. Nothing holds it anywhere. Unconnected, it picks up charge from the air, a finger, the rod next to it, and wanders through the forbidden band. An uninitialised variable.

A pull resistor fixes it: a weak tie to supply or ground that sets the default, which any real driver overrides. Because the leakage is nanoamps, even 100 kΩ holds firm: 10 nA × 100 kΩ = 1 mV.



A digital input is a voltage measurement, and an unconnected pin measures noise. Pull resistors give every input a default. The three in this build: 10 kΩ holds GPIO2 high because A1 reads it during reset to choose how to boot; 100 kΩ keeps the mic's data line from floating between frames; the amp's SD pull-down keeps the amp off (below 0.08 V) until A1 is awake and drives it. The button uses A1's built-in pull-up.

The four defaults in this build

GPIO2 is a strapping pin. Out of reset, the chip reads it to decide how to boot, and it must read high. Nothing is driving yet, so 10 kΩ to 3.3 V answers.

GPIO4 is the mic's data line. The mic drives it during its slot and lets go the rest of the time. 100 kΩ to ground stops it floating in between.

GPIO5 is the amp's shutdown input: below 0.08 V, off. The pull-down guarantees that while A1 resets and GPIO5 floats, so the speaker never pops at power-up.

The button uses A1's internal pull-up. High until pressed to ground.

Push-pull versus open-drain

Push-pull drives a line hard to either rail; I²S works this way. Open-drain can only pull low and needs a pull-up for the high. Open-drain lets many devices share a wire without fighting, which is why I²C uses it.

The price is speed. The rising edge is chapter 4's charging curve, $\tau = R_{pullup} \times C_{line}$. 10 kΩ into 50 pF is half a microsecond per edge. So I²C tops out at a few hundred kilohertz while push-pull I²S runs at a megahertz without trying.

> IN THE BRASS BUILD, THIS IS WHY
>
> - 10 kΩ on GPIO2 to the 3V3 rail: a boot strap must read high before anything drives it
> - 100 kΩ on GPIO4 to the GND rail: the mic releases the line between frames
> - the amp's SD pull-down to the GND rail: off through reset
> - no resistor on the button; A1's internal pull-up does it

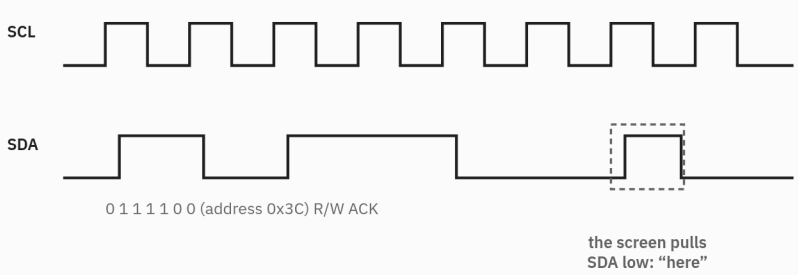
CHAPTER 6 OF 11

I²C: a bus with addresses

Two wires, many devices. SCL is a clock, SDA is data, both open-drain with pull-ups, so they idle high and anyone can pull them low.

The controller starts a transaction by pulling SDA low while SCL is high. It clocks out a 7-bit address plus a read/write bit. Every device listens; the one that matches pulls SDA low on the ninth clock. That is the acknowledge. Bytes flow, each acknowledged. SDA rising while SCL is high ends it.

Data may only change while SCL is low. That one rule is what makes start and stop distinguishable from data.



Bits change while the clock is low and are read while it is high. 100–400 thousand times a second: slow, by chip standards.

I²C is a two-wire bus with addresses, like a tiny network. A1 is the only talker; it clocks SCL and puts the address 0x3C (or 0x3D) on SDA; the screen answers. Nobody drives the lines high: devices only pull them low, and resistors on the screen module pull them up (chapter 4 again). At a few hundred kilohertz, 30 mm of brass rod is electrically a perfect wire, so the rods can run wherever is tidy.

Your screen on the bus

The SSD1306 answers at 0x3C or 0x3D, set by a solder bridge on the module. The firmware tries both.

It holds 128 × 64 pixels, one bit each: 1024 bytes. A full frame at 400 kHz, with overhead, is about 25 ms. Forty frames a second at best. The bus, not the processor, is why the UI is simple.

Why bare rods don't care

At 400 kHz a bit lasts 2.5 µs. In that time a signal travels 750 m. The rod is 30 mm. Its length and shape mean nothing to the signal; it adds about a picofarad to the line.

The only constraints on these two rods are mechanical: touch nothing.

> IN THE BRASS BUILD, THIS IS WHY
>
> - two rods, SCL and SDA, routed wherever is tidy
> - no extra pull-ups: the screen module carries its own
> - the firmware probes both addresses because the module's solder bridge picks one

CHAPTER 7 OF 11

Sound into numbers: sampling and I²S

A microphone is a moving capacitor

Sound is air pushed slightly denser, then thinner, hundreds to thousands of times a second. The mic's diaphragm, a silicon membrane a millimetre across, moves with it.

In a MEMS mic the membrane is one plate of a capacitor and a fixed perforated plate is the other. Motion changes the capacitance. A tiny amplifier and an analogue-to-digital converter on the same chip turn that into numbers. Nothing analogue ever leaves the package.

Sampling

Measure the wave at regular instants and you capture everything up to half the sampling rate. Anything above folds back as a false lower tone, aliasing, so the converter filters it out first.

16 000 samples a second hears up to 8 kHz. Phone speech is cut at 3.4 kHz and still intelligible; 8 kHz keeps consonants crisp. Music wants 44.1 kHz, but this is a voice device and fewer samples means fewer bytes over Wi-Fi.

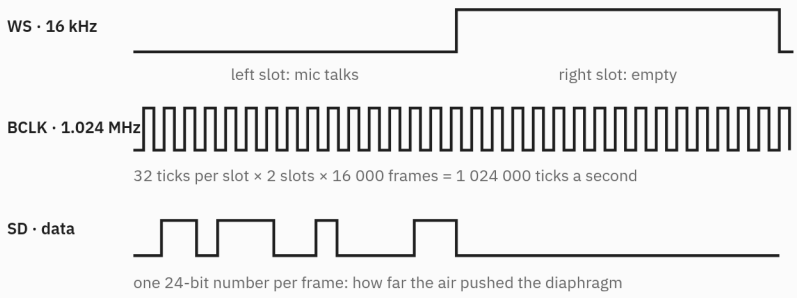
Each sample is a 24-bit integer: 144 dB of range on paper. The mic's own noise limits it to about 61 dB. Still a quiet room to a shout.

I²S is a stream, not a bus

Three wires. WS flips once per sample and marks left and right slots. BCLK ticks once per bit. SD carries the bits. With 32-bit slots and two channels: 16 000 × 2 × 32 = 1.024 MHz.

A1 generates both clocks. The mic and the amp both follow them, so sharing WS and BCLK between the two is natural. One rod, two taps.

The mic speaks only in the slot its L/R pin selects. The firmware reads the left slot, so L/R goes to ground. Wire it the other way and samples count up while every value is zero: the mic is talking into the slot nobody reads.



16 000 samples a second captures sound up to 8 kHz, which covers speech.
The mic and the amp share WS and BCLK: one rod each, two taps.

I²S is the same idea at a thousand times the speed, with a clock the firmware chose. A1 generates both clocks, so sharing them between mic and amp is natural: that is why those two rods have two taps. The mic's L/R pin wired to GND makes it speak in the left slot, the one the firmware reads; get that wrong and you see samples counting with min and max stuck at 0. At 1 MHz the wavelength is 300 m, so a 40 mm rod is still electrically short, but this is the fastest signal in the box, so it is the one to keep away from the antenna corner.

Frequency, wavelength, and a 40 mm rod

λ = v / f. At 1 MHz on a conductor, λ is 300 m. The rod is 40 mm: electrically a point. The edges are a few nanoseconds, roughly 100 MHz, still 3 m. Rule of thumb: under a tenth of a wavelength, a conductor is just a node.

So the rods are fine for the signal. Their one weakness: a fast edge radiates a little, and the most sensitive thing in the box is the antenna.

RMS

Samples swing around zero, so their average is meaningless. Square them, average, square-root: RMS, one number for how loud the last second was. The fast-track test asks for quiet and speech RMS because the *change* is the proof, not a threshold.

> **IN THE BRASS BUILD, THIS IS WHY**
>
> - WS and BCLK rods each have two taps
> - the mic's L/R link to the GND rail picks the left slot
> - the fastest signals in the box stay away from the antenna corner

CHAPTER 8 OF 11

Making sound: class D, the bridge, decibels

Switches, not a volume knob

A speaker coil needs real current, far more than a logic pin. A class-D amp supplies it by switching its outputs fully on and off a few hundred thousand times a second and varying the on-time, so the *average* traces the audio.

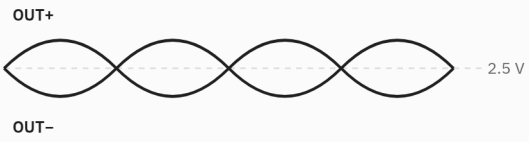
Switches fully on or off waste almost nothing. That is why a fingernail-sized chip drives a speaker with no heatsink. The coil is an inductor; it smooths the switching and the cone follows the average. The scope needed a 20 kHz low-pass for the same reason.

The bridge

The MAX98357A has two outputs, OUT+ and OUT−, driven in opposite directions, with the speaker between them. Each swings nearly 5 V; the speaker sees the difference, up to twice that. Four times the power of a single output from the same supply.

The price: neither output is ground. Both sit near 2.5 V with no signal. Connect either to ground, or to a grounded frame, and you have shorted half the bridge.

Bridge-tied load: the speaker hangs between two live outputs



the speaker sees OUT+ minus OUT−: twice the swing from one 5 V supply

neither output ever sits at 0 V → neither may touch GND or the brass

Volume cap (firmware)

```
cap kept peak @15 dB
15  2 %  0.24 V
30  9 %  0.98 V
45 20 %  2.20 V ✔
50 25 %  2.71 V ✘
```

speaker limit 2.53 V (0.8 W into 8 Ω)

Class-D and the bridge. The amp switches its outputs at a few hundred kHz and the speaker's coil smooths that into sound. Driving the speaker between two outputs doubles the swing, which is why a 5 V part is loud, and why both speaker leads are live. The firmware squares the cap before applying it, so cap 45 passes only 20 % of full amplitude: 2.20 V peak even at the 15 dB gain your 100 kΩ GAIN resistor selects, under the speaker's 2.53 V rating.

The volume cap, derived

P = V_rms² / R. The speaker is 8 Ω, 0.8 W, so V_rms max = √(0.8 × 8) = 2.53 V. The worst case is a clipped, square-like signal whose peak equals its RMS. So the limit is 2.53 V *peak*: a square wave there delivers 0.8 W, a sine only 0.4 W.

The firmware squares the cap before applying it. Loudness perception is roughly logarithmic, and a squared slider feels evenly spaced. Cap 45 → 0.2025 of full scale. Full scale at your 15 dB gain is about 10.9 V peak, so cap 45 → 2.2 V. Under the line at any gain.

Cap 50 → 2.7 V. Over. That is why the shortcut stops at 45 and anything higher goes through the dummy load and the scope.

Decibels

A decibel is a ratio. Power: 10 $\log_{10}(P/P_0)$. Amplitude: 20 $\log_{10}(V/V_0)$, because power goes as V^2. Every 6 dB doubles an amplitude; every 10 dB is ten times the power.

The GAIN pin's 9, 12, 15 dB are amplitude ratios of 2.8, 4, 5.6. Cap 45 is 20 log(0.2025) ≈ −14 dB. And 8.5 dBm is 8.5 dB above a milliwatt: 7 mW, against the default 20 dBm, 100 mW.

> IN THE BRASS BUILD, THIS IS WHY
>
> - OUT+ and OUT− never touch GND, a rail or the frame
> - the 100 kΩ GAIN resistor to the GND rail sets 15 dB; the cap stays at 45 until the scope clears more
> - the speaker's own leads go to the amp's terminal block, never cut or stripped

CHAPTER 9 OF 11

The antenna and the near field

Fields that feed each other

A changing electric field makes a magnetic field; a changing magnetic field makes an electric one. Maxwell showed the pair can sustain itself and travel at the speed of light. λ = c / f: at 2.45 GHz, 122 mm.

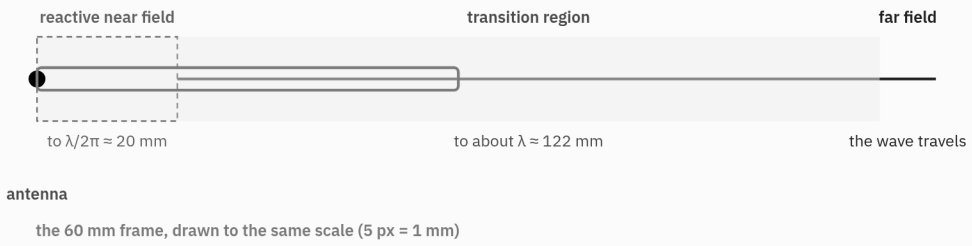
An antenna is a conductor shaped to launch that wave. It works best when resonant, usually a quarter wavelength: 30 mm here. A1's chip antenna is far shorter because its ceramic shrinks the wavelength inside it.

It is also only half the antenna. The board's ground copper carries the mirror current. The SuperMini's problem is that copper sits too close to the chip, and the two halves interfere.

Close in, the wave hasn't left yet

Within about λ/2π, 20 mm, energy sloshes between the antenna and whatever is near it each cycle. This is the reactive near field, and it is where metal does damage.

Any conductor there joins the resonant system. It adds capacitance and shifts the resonance away from 2.45 GHz. It carries induced currents that re-radiate elsewhere. If it is lossy it turns signal into heat. None of this needs contact. Tape stops a short; it does nothing about a field.



Three regions around any antenna. Close in, energy sloshes back and forth between the antenna and whatever is near it; that is where metal does the most damage. Espressif's 15 mm clearance figure sits right inside this zone, which is why it is a starting rule rather than a guarantee.

Mismatch

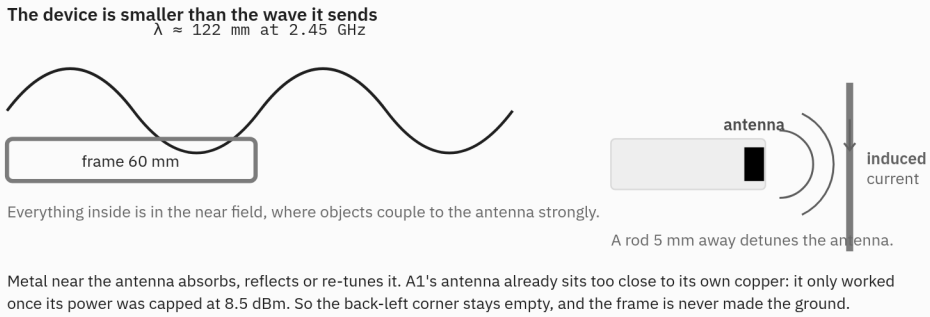
The radio feeds the antenna through a matching network tuned for the antenna's expected impedance. Detune the antenna and part of the power reflects back instead of leaving.

Your three boards' hotspot was invisible at 20 dBm and appeared at 8.5 dBm. Nobody has proven why. Either the power amplifier distorts into the mismatched load or the board's little regulator sags under a 100 mW burst. Either way: this antenna has no margin, and the enclosure must take none.

And the frame?

A grounded brass cage would be a large conductor at fixed potential inside the near field: a ground plane and a partial shield at once. It might help one direction and kill another. An isolated frame is at least not tied to the circuit; it still sits in the field.

So: antenna end at an outside edge, air above it, rods and rails on the far side of the box. Then the only instrument that can answer the question: signal strength inside the frame versus the open bench, several runs each, because a hand's movement shifts RSSI by several dB.



Metal near the antenna absorbs, reflects or re-tunes it. A1's antenna already sits too close to its own copper: it only worked once its power was capped at 8.5 dBm. So the back-left corner stays empty, and the frame is never made the ground.

Why a corner of the box stays empty. At 2.4 GHz the antenna's field fills the whole frame, and any conductor in it becomes part of the antenna. Grounding the frame would wrap the antenna in a ground plane; running a rail past it would add a parasitic element. The last step of the build, comparing Wi-Fi inside the frame with the open bench, is the only real test; the clearance rules just give it the best chance.

> IN THE BRASS BUILD, THIS IS WHY
>
> - the back-left corner stays empty with nothing above it: that is the reactive near field
> - the frame is never ground
> - the Wi-Fi comparison at the end is the only proof; the rules just give it the best chance

CHAPTER 10 OF 11

Bare conductors: when they work and when they don't

Two rods side by side are a capacitor

Two 1 mm rods 3 mm apart: about 16 pF per metre, so 0.6 pF over 40 mm. A 1 MHz edge pushes a nanoamp-scale blip through that into its neighbour. That is crosstalk, and at these speeds it is nothing.

Insulation would not change that capacitance. It only stops the rods touching. Electrically, the insulated plan was never needed.

Touching is the entire risk

Two signal rods touch: two push-pull drivers fight, one high, one low, tens of milliamps through each other until one overheats.

A signal rod touches the 3V3 rail: a pin straight to the supply. The protection diodes may survive it briefly. May not.

The 5 V rod touches anything on the 3.3 V side: over the absolute maximum of every chip there.

Any rail touches the frame: the frame now has a potential, and the next accidental touch anywhere on the cage completes a loop.

Every one of these is a few milliohms of brass from a healthy circuit. Hence three habits: sleeve crossings, stay 2 mm from the frame, meter after every rod while the fault is one rod old.

The mechanical bonus

A 1 mm rod is hundreds of times stiffer than 30 AWG wire. Soldered, the harness is a rigid truss; the modules are held by their own connections and nothing flexes a joint when you handle the device. Joints fail from repeated bending, not from standing still.

The trade: a drop sends the shock straight into the joints. One more reason to keep the harness small and the joints properly wetted.

> IN THE BRASS BUILD, THIS IS WHY
>
> - sleeves at every crossing and 2 mm from the frame: remove the only failure mode
> - a meter check after every rod: GND rail to 3V3 rail, to the 5 V rod and to the frame must never beep
> - rods are electrically equal to wire and mechanically better; the cost is care

CHAPTER 11 OF 11

Heat, solder and flux

Temperature is a level, heat is a flow

Heat flows from hot to cold. How much a part needs to warm up scales with its mass; how fast it can get it is conduction. Brass conducts well, 110 W/m·K, and a frame loop weighs far more than a header pin. Touch the iron to the frame and the heat spreads into the whole loop; the joint sits below melting for seconds.

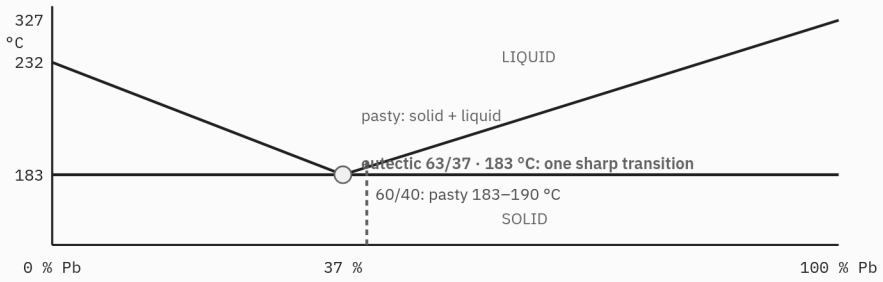
A chip pin is tiny and the board under it conducts at 0.3 W/m·K. The pin is at temperature almost at once, and so, soon after, is the chip. The chip is rated for roughly 260 °C for ten seconds.

So: 370–400 °C and a broad tip for the frame, to beat the heat soak. 330–350 °C, a fine tip, and one or two seconds for a module pin.

Why 63/37

Tin melts at 232 °C, lead at 327 °C, mixtures lower. At 63 % tin, 37 % lead the alloy melts like a pure metal: solid to liquid at one temperature, 183 °C. That mix is the eutectic.

Any other mix melts over a range, and inside that range it is slush, solid crystals in liquid metal. Jog a joint in the slush and it freezes grainy and cracked: a cold joint that fails months later. 63/37 has no slush. Hold it still five seconds and you get a shiny fillet every time.



Why your solder is 63/37 and not 60/40. At the eutectic mix the alloy freezes at a single temperature. Any other mix passes through a pasty range where a nudge makes a grainy, cracked joint. The joint you hold still for five seconds is passing through 183 °C.

A joint is a chemical bond, not glue

Molten solder on clean metal wets it, spreading in a film pulled by surface tension. At the boundary the tin reacts with the base metal and forms an intermetallic layer a few micrometres thick. That layer is the bond. A good joint conducts like the metals on either side.

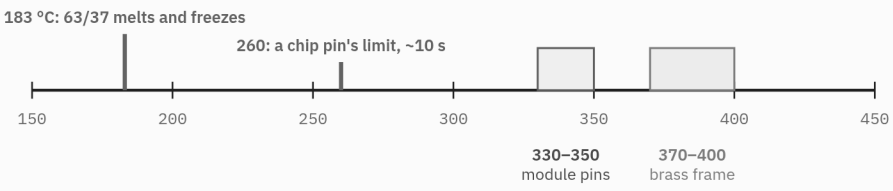
Wetting fails on oxide. Every metal grows an oxide skin in air within minutes, brass faster than copper, and solder balls up on oxide like water on wax. Flux is an acid that wakes up at soldering temperature, dissolves the oxide, and keeps air off until the solder lands.

Two fluxes, two zones

The Chip Quik pen is rosin-based: mild organic acids, enough for copper and freshly sanded brass. Its residue is a resin that neither conducts nor corrodes, so it can stay on a board.

The Harris flux is zinc chloride in water. Strong enough to strip unsanded brass, which is why frame joints flow so easily. Its residue is a salt that pulls moisture from the air. Damp salt conducts. A film of it across two I²S pads leaks microamps that no meter finds and no debugging explains, and it keeps eating the metal underneath.

So: acid flux on the empty frame only. Washed the same day. Neutralised with baking soda, a mild base that turns leftover acid into harmless salts. Alcohol last, to push the water out so nothing sits wet.



The iron runs far above the solder's melting point because the joint must reach 183 °C quickly, before heat soaks into the part. Brass is a thick heat sink, so it gets the hotter tip and the hotter setting; a module pin gets 1–2 seconds and no more.

Soldering is a race between the joint and the part. 63/37 is a eutectic alloy: it goes from solid to liquid at one temperature, 183 °C, with no pasty range, so joints that cool undisturbed come out shiny. Flux strips the oxide that stops solder wetting metal. The Harris flux does that with an acid strong enough for brass, and its residue conducts and corrodes, which is why it never comes near a module: a few microamps of leakage on an I²S line is a fault you would never find.

IN THE BRASS BUILD, THIS IS WHY

- brass joints: 370–400 °C, broad tip, Harris flux, then hot water, baking soda and alcohol the same day
- module joints: 330–350 °C, fine tip, Chip Quik flux, one to two seconds, rod ends sanded first
- the brass tip and the acid flux never touch a module

Check yourself

A friend says the rods must be slower than wire because brass is a worse conductor. What do you tell them?

Material resistivity is four times copper's, but the rod's cross-section is sixteen times the wire's, so the rod is about four times lower in resistance: 3 mΩ against 14 mΩ over 40 mm. Either is a millivolt at 0.3 A.

Why does the 220 µF capacitor have to be at the amplifier and not, say, at A1 where there is room?

Every millimetre of conductor is about a nanohenry, and V = L di/dt. A class-D edge of 0.3 A in 30 ns through 300 nH of path drops 3 V. The capacitor's job is to make the loop a few millimetres long, which only works next to the pins.

The mic's samples count up but every value is zero. Which chapter explains it?

Chapter 7. The mic speaks in the slot its L/R pin selects; with L/R not grounded it talks in the right slot and the firmware reads the left. Nothing is broken.

Why can cap 45 go straight to the speaker but cap 50 cannot?

The speaker's 0.8 W into 8 Ω means 2.53 V peak for a worst-case square wave. Cap 45 squared is 0.2025 of full scale, 2.2 V at the highest gain; cap 50 is 0.25, 2.7 V. Over the line, so it must be measured on the dummy load first.

Kapton tape separates A1 from the brass. Is the antenna safe?

No. Tape prevents contact, not coupling. Within λ/2π of the antenna, 20 mm, any conductor adds capacitance and detunes it. Only the inside-versus-bench Wi-Fi test tells you what the frame did.

Why neutralise the frame with baking soda instead of just rinsing?

Zinc chloride residue is an ionic salt that keeps attracting water and keeps corroding; rinsing removes most but not the film in the joints. Bicarbonate is a base that converts the acid to inert salts, which then rinse away.

Parts mentioned

PRODUCT YOU BOUGHT	CHAPTERS	LINK
Meshnology ESP32-C3 SuperMini (A1)	1, 3, 5, 9	B0F888JQ91
K&S #9861 1.0 mm brass rod	2, 10	B005WPB7YG
LuminologyPro resistor kit	2, 3, 5, 8	B0F4P352BB
ALLECIN electrolytic (220 µF) and BOJACK ceramic (100 nF) kits	4	owned · no link recorded
Hosyond SSD1306 OLED	6	B09T6SJBV5
AITRIP INMP441 microphone	7	B092HWW4RS
HiLetgo MAX98357A amplifier	4, 8	B0CDWXZZCH
Same Sky CES-20134-088PM speaker	8	DigiKey
KAIWEETS HT118A multimeter	4, 10	B08BL288LW
MAIYUM 63/37 solder · Harris SCLF4 flux · Chip Quik CQ4LF pen	11	B076QF1Y85 · B0015DWPV8 · owned
ELEGOO Kapton tape · Pointool heat-shrink	9, 10	B072Z92QZ2 · B08N4W4K9X

Read deeper

- In the repository: `edu/fundamentals/` lessons 01–02 (units, DC circuits), 03 (components), 06 (power integrity), 07 (GPIO and boot straps), 08 (I²C), 09 (I²S), 10 (class-D and acoustics), 11 (RF), 12 (soldering and tolerance); and the short notes in `plan/concepts/`.
- Primary sources the numbers come from: the ESP32-C3 datasheet (logic thresholds, strapping pins, transmit power), the MAX98357A datasheet (shutdown threshold, gain, full-scale output), the INMP441 datasheet (sensitivity and SNR), and Espressif's PCB layout guide (antenna clearance).

Sources. Device numbers are the project's: the 1.024 MHz bit clock, the 0.08 V shutdown level, the squared volume cap and the 2.53 V limit from `plan/FAST_TRACK.md`; the 8.5 dBm cap from `docs/INVENTORY.md`. Logic thresholds (0.25 and 0.75 × VDD) are the ESP32-C3 datasheet's; resistivities, thermal conductivities, the Sn-Pb eutectic and the 260 °C / 10 s rating are handbook values. The 50 Ω driver resistance, 1 nH/mm, 50 pF line and the ≈ 16 pF/m rod capacitance are engineering estimates for illustration, not measurements.

What is and is not verified. Everything here is explanation, not test evidence. Where the text says the cause of the SuperMini's antenna problem is not settled, that is the honest state of knowledge in the project. If this page and the repository differ, the repository wins.